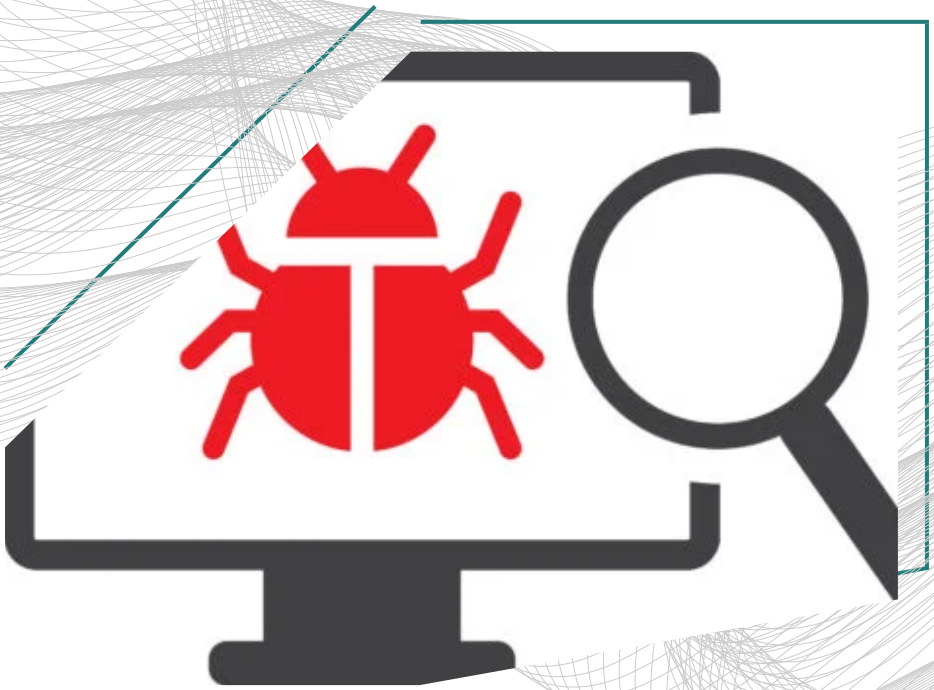
CYBERIUM

BUG BOUNTY



INTRODUCTION

The Bug Bounty Hunting Course is designed to equip learners with the skills required to identify, exploit, and report security vulnerabilities in web applications. As organizations increasingly rely on digital infrastructure, ethical hackers play a crucial role in helping companies secure their systems by identifying security flaws before malicious hackers can exploit them. This course provides a structured learning path for individuals looking to enter the bug bounty and ethical hacking industry, offering practical insights into hacking methodologies, vulnerability identification, and responsible disclosure practices.

Through a combination of theoretical knowledge and hands-on labs, participants will gain expertise in web application security testing, network security, and API exploitation. The course also introduces popular bug bounty platforms like HackerOne, Bugcrowd, and Open Bug Bounty, helping learners understand how to legally and ethically earn rewards by finding security vulnerabilities. By the end of this course, participants will have the skills to analyze, exploit, and report vulnerabilities effectively, making them valuable assets in the cybersecurity industry.

COURSE OBJECTIVES

By completing this course, participants will be able to:

- Understand bug bounty programs, ethical hacking principles, and responsible disclosure.
- Identify and exploit OWASP Top 10 vulnerabilities in web applications.
- Conduct reconnaissance and information gathering to discover hidden security flaws.
- Perform network and API security testing to uncover vulnerabilities in cloud services and web applications.
- Gain hands-on experience with bug bounty tools like Burp Suite, Nmap, Metasploit, Sublist3r, and Amass.
- Learn report writing techniques to submit high-quality bug reports and maximize earnings.
- Get familiar with bug bounty platforms (HackerOne, Bugcrowd, Intigriti, Synack, Open Bug Bounty) and how to participate in them.

BUG BOUNTY SYLLABUS

Module 01- Introduction to Bug Bounty Hunting

1. Understanding Bug Bounty Programs and Responsible Disclosure
2. Differences between Ethical Hacking, Penetration Testing, and Bug Bounties
3. Overview of popular Bug Bounty Platforms (HackerOne, Bugcrowd, Intigriti, Synack, Open Bug Bounty)
4. Legal and ethical considerations for bug bounty hunting

Module 02- Setting Up the Bug Bounty Lab

1. Installing Kali Linux, and Burp Suite
2. Configuring Virtual Machines and Cloud-Based Environments
3. Using Proxies, VPNs, and Anonymity Tools for ethical hacking
4. Best practices for creating a safe and legal testing environment

Module 03- Reconnaissance and Information Gathering

1. Passive vs. Active Reconnaissance Techniques
2. DNS Enumeration with Amass, Sublist3r, and Recon-ng
3. Discovering subdomains and exposed services
4. Google Dorking, Shodan, and OSINT (Open-Source Intelligence) techniques



Module 04- Web Application Security Testing

1. Understanding OWASP Top 10 Vulnerabilities
2. Using Burp Suite for Manual Web Application Testing
3. Automating scans with Nikto, OWASP ZAP, and Wfuzz

Module 05- SQL Injection

1. Understanding SQL Injection and its impact
2. Types of SQLi: Error-Based, Union-Based, Blind (Boolean & Time-Based)
3. Exploiting SQLi with SQLmap and Manual Techniques
4. Bypassing WAF (Web Application Firewall) Protections
5. Preventing SQL Injection
6. Hands-on Lab: Exploiting SQLi

Module 06- Broken Authentication

1. Understanding Authentication Flaws and their consequences
2. Exploiting weak authentication mechanisms: Brute Force, and Session Hijacking
3. Bypassing authentication using SQL Injection, Default Credentials, and Cookie Tampering
4. Multi-Factor Authentication (MFA) Bypass Techniques
5. Preventing Broken Authentication: Secure Session Management, Strong Password Policies, and Account Lockout Mechanisms
6. Hands-on Lab: Exploiting and Fixing Authentication Vulnerabilities

Module 07- Broken Access Control

1. Understanding Access Control Failures
2. Exploiting IDOR (Insecure Direct Object References)
3. Privilege Escalation via Role Manipulation
4. Preventing Broken Access Control
5. Hands-on Lab: Finding and Exploiting Broken Access Control Vulnerabilities

Module 08- File Upload Vulnerabilities

1. Understanding File Upload Risks (Remote Code Execution, Privilege Escalation)
2. Exploiting Unrestricted File Upload to Gain Server Access
3. Bypassing File Type Restrictions
4. Preventing File Upload Vulnerabilities
5. Hands-on Lab: Uploading a Malicious Shell and Gaining Remote Access

Module 09- Command Injection

1. Understanding OS Command Injection and its impact
2. Exploiting Command Injection using Shell Commands and Pipes
3. Bypassing Input Filters and Restrictions
4. Real-World Command Injection Exploits
5. Preventing Command Injection: Input Validation, Least Privilege Execution, and Safe API Calls
6. Hands-on Lab: Executing Remote Commands on a Vulnerable System

Module 10- Path Traversal (Directory Traversal)

1. Understanding Path Traversal Attacks and their impact
2. Exploiting Path Traversal to Access Restricted Files (passwd, config, logs)
3. Using ../, URL Encoding, and Null Byte Injection to Bypass Filters
4. Preventing Path Traversal
5. Hands-on Lab: Extracting Sensitive Files from a Vulnerable Web Application

Module 11- Information Disclosure

1. Understanding Information Leakage Risks
2. Identifying Exposed Sensitive Data in Error Messages, and Public Repositories
3. Exploiting Misconfigurations
4. Preventing Information Disclosure
5. Hands-on Lab: Extracting Sensitive Data from Exposed Services

Module 12- Cross-Site Request Forgery (CSRF)

1. Understanding CSRF and its consequences
2. Exploiting CSRF to Perform Unauthorized Actions on Behalf of Users
3. Crafting Malicious Requests in HTML, JavaScript, and iFrames
4. Preventing CSRF
5. Hands-on Lab: Creating a Malicious CSRF Exploit and Fixing it

Module 13- Cross-Site Scripting (XSS)

1. Understanding XSS Attacks and their impact
2. Types of XSS: Stored, Reflected, and DOM-Based
3. Exploiting XSS to Steal Cookies, Deface Web Pages, and Conduct Phishing Attacks
4. Bypassing XSS Filters Using Encoding, Obfuscation, and Event Handlers
5. Preventing XSS
6. Hands-on Lab: Injecting and Executing Malicious Scripts in a Web Application

Module 14- OAuth Vulnerabilities

1. **Understanding OAuth Authentication and Authorization Flaws**
2. **Exploiting Improper Redirects, Token Leakage, and Insufficient Scopes**
3. **Real-World OAuth Attacks**
4. **Preventing OAuth Exploits**
5. **Hands-on Lab: Identifying and Exploiting an OAuth Misconfiguration**

Module 15- Report Writing

1. **Writing detailed bug reports with PoC (Proof of Concept)**
2. **Applying all techniques learned in real-world bug bounty scenarios**
3. **Hunting for vulnerabilities in open-source applications and live bug bounty programs**
4. **Submitting findings to HackerOne, Bugcrowd, and Intigriti**

Note: Please confirm payment details via our official WhatsApp +91-9318492128 before making any payment. Ensure the Payment Account name is 'CyberiumX' only.